Fiscal Year:	FY 2026	**Task Last Updated:**	FY 06/25/2026
PI Name:	Taylor, Christopher Ph.D.		
Project Title:	Hydroponic Microbes in a Low Carbon, High-Radiation Environment; Identifying the Genetic Determinants of Water- and Plant-Associated Microbial Survival in Extraterrestrial Environments		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2024 Space Biology NNH24ZDA001N-SBR: E.9 Space Biology Research Studies
Start Date:	04/01/2026	**End Date:**	03/31/2030
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA HQ
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Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Benitez Ponce, Maria Ph.D. (Ohio State University) Newman, Adam Ph.D. (Ginkgo Bioworks, Inc.)		
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Task Description:	Main Objective and Specific Aims: Microbes play a crucial role in plant survival by enhancing growth, health, and stress resilience, particularly in resource-limited environments. This research aims to identify key genetic pathways that govern the survival of hydroponic- and plant-associated microbes in extraterrestrial regolith (e.g., Moon, Mars), with implications for sustaining long-term plant, animal, and human survival during missions to the Moon (Artemis/Gateway) and Mars. The specific aims for the proposed research are: ● Assess the survival of hydroponic- and plant-associated bacteria in different synthetic model regoliths (Moon/Mars) under cosmic ray-simulated radiation. ● Evaluate microbial survival and community dynamics in synthetic microbiomes. ● Determine plant-microbe interaction outcomes under oxidative stress. Project Description: Plant cultivation in space, both now and in the future, will likely continue to rely on a combination of hydroponics for nutrient delivery and/or regolith as a plant support medium. Modern hydroponic systems must navigate microbial communities that range from beneficial, promoting plant growth; to commensal, having neutral effects; to harmful, posing risks to plant health. Ongoing research seeks to understand the function of hydroponic-associated microbiomes and develop strategies to engineer them for improved disease resistance, nutrient acquisition, and stress tolerance. Recent advancements in mass DNA sequencing, genome assembly, and bioinformatics have revolutionized the study of micro- and macro-organismal genomes that frequent hydroponic systems, providing researchers with an unprecedented opportunity to identify genes that influence fitness and survival and enhance overall plant health. Our research aims to identify genetic pathways in plant- and hydroponic-associated microbes that enhance microbial survival and adaptation in low-carbon, high-radiation environments--conditions expected in hydroponic production systems in space and on the Moon and Mars regolith. Comparative genome analyses of microbes across species enable the identification of genes that drive diversity and confer selective advantages in extreme environments. Using bacteria as a study system, this project will examine two critical aspects of hydroponic and plant-associated microbial adaptations in response to space conditions: (1) survival under radiation, and (2) colonization of regolith and their impact on hydroponic system-associated functions, such as colonization, cross-protection to space stresses, overall impact on plant growth. To address these questions, we have assembled a team of public- and private-sector researchers with extensive expertise in microbial library development and experimentation, microbial identification, comparative genomics, bioinformatics, and a wide range of experimental studies on microbial impacts on plant health, including microbiome assemblage, nutrient acquisition, and pathogen suppression. Given the extensive microbial libraries and genomic resources available, this research presents a unique opportunity to uncover genetic mechanisms that enhance microbial survival in space. These findings will inform future efforts to develop resilient microbiomes capable of withstanding the challenges of space travel while continuing to support plant, animal, and human life in extraterrestrial environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Invariable food production systems in space will be contaminated with terrestrial microbes, and, due to the harsh conditions of space, unexpected outcomes (e.g., failure, human pathogen colonization) may occur as a result. To gain a better understanding of the factors governing microbial survival in space, we will examine a wide range of different microbes known to interact with and frequent hydroponic systems. By screening very large numbers of bacterial strains and utilizing available genomic information, we will be able to establish correlations between survival and specific genes and pathways. The identification of survival mechanisms has the potential to facilitate the selection or engineering of microbes with improved performance – not only for plants, but also for humans and other biological systems employed in space research.
Task Progress:	New Project for FY2026
Bibliography Type:	Description: (Last Updated:)